The Capability Trap: Why Symbolic Consensus Cannot Beat Majority Voting for LLM Reasoning

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Abstract

Recent work proposes augmenting Large Language Model (LLM) reasoning with symbolic methods: extracting formal constraints from reasoning traces and using satisfiability solvers for answer aggregation. We investigate when such methods outperform simple majority voting (Self-Consistency, SC). We instantiate symbolic consensus as Sica (Symbolic Invariant Consensus Aggregation), which extracts first-order logic (FOL) constraints from sampled reasoning traces and aggregates them via weighted MAX-SAT. Across 5 models spanning three architecture classes (instruction-tuned, distilled-reasoning, full-reasoning), 6 reasoning domains, and over 1,300 test problems, we find that Sica provides no significant accuracy gain over SC. The accuracy gap ranges from -3.3 to $+5.0$ percentage points across conditions, with no result reaching $p < 0.10$. A constraint ablation reveals the mechanism: MAX-SAT, simple constraint counting, and random-weight aggregation agree on 99.4% of problems, confirming that extracted constraints lack discriminative power. We term this the *capability trap*: instruction-scale models produce diverse outputs (multi-candidate rate 38–62%) but non-discriminative constraints (precision ≈ 0.54), while reasoning models eliminate diversity entirely ($C_0 < 5\%$). We propose C_0 (the fraction of problems with multiple candidate answers) as a zero-cost binary gate: when $C_0 < 5\%$, symbolic consensus degenerates to majority voting and can be skipped.

1 Introduction

Sampling multiple reasoning traces from an LLM and aggregating their answers via majority voting, known as Self-Consistency (SC), is a simple and effective strategy for improving accuracy on reasoning tasks. A natural question is whether more sophisticated aggregation can do better. Several recent methods propose extracting symbolic representations (formal constraints, logic programs, verification conditions) from reasoning traces and using solvers to select answers. The intuition is appealing: formal reasoning should catch errors that voting misses.

But when does this actually help? Despite growing interest in neuro-symbolic reasoning pipelines, no prior work systematically characterizes the conditions under which symbolic aggregation outperforms voting. Individual papers report gains on specific benchmarks, but the interaction between model capability, output diversity, and constraint quality remains unexplored.

We address this gap with a large-scale empirical study. We instantiate symbolic consensus as Sica (Symbolic Invariant Consensus Aggregation), a method that extracts FOL constraints from each sampled reasoning trace and selects the answer that satisfies the maximum weighted set of constraints via a MAX-SAT solver. We evaluate Sica against SC across 5 models (Qwen-7B/14B/72B, LLaMA-8B, QwQ-32B), 6 reasoning domains (ProofWriter, FOLIO, StrategyQA, ARC-Challenge, GSM8K, AIME), and over 1,300 problems.

The results are uniformly negative. No model-domain combination yields a significant accuracy gain for Sica over SC. More importantly, a constraint ablation experiment reveals *why*: MAX-SAT optimization over the extracted constraints produces identical answers to simple constraint counting on 99.4% of problems (801/804). The solver's satisfiability reasoning contributes

nothing because the constraints themselves lack discriminative power, with mean precision near chance at 0.54.

We identify a structural explanation that we term the *capability trap*. Symbolic consensus requires two ingredients: (1) output diversity, so there are multiple candidate answers for the solver to adjudicate, and (2) discriminative constraints, so the solver can correctly distinguish answers. These requirements are empirically incompatible across the tested conditions and model scales. Instruction-scale models produce abundant diversity (multi-candidate rate 38–62%) but their constraints are near-random. Full reasoning models produce high-quality reasoning chains but eliminate diversity entirely (multi-candidate rate $< 5\%$), making symbolic aggregation degenerate to voting.

Our contributions are:

1. An empirical demonstration that symbolic consensus (instantiated as SICA) provides no significant accuracy gain over majority voting across 5 models, 6 domains, and $N > 1{,}300$ problems. This limitation persists across the tested temperature range (0.3–1.0) and ensemble sizes ($K = 6$–20).
2. A mechanistic explanation via the *capability trap*: MAX-SAT $\approx$ constraint counting $\approx$ random weighting (99.4% agreement), because extracted constraints lack discriminative power (precision ≈ 0.54) regardless of output diversity level.
3. A practical diagnostic: C_0 (multi-candidate rate) serves as a zero-cost binary early-stopping gate. When $C_0 < 5\%$, symbolic consensus degenerates to SC and practitioners can skip the overhead.

2 Method: Symbolic Invariant Consensus Aggregation

2.1 Overview

Given a question q, SICA operates in four stages: (1) sample K reasoning traces from an LLM, (2) extract FOL constraints from each trace, (3) deduplicate and weight constraints, and (4) select the answer via weighted MAX-SAT.

2.2 Trace Sampling

We sample K independent reasoning traces $\{t_1, \ldots, t_K\}$ from the LLM using temperature T. Each trace produces a candidate answer a_i. The set of distinct candidate answers defines the candidate pool $\mathcal{A} = \{a_i\}_{i=1}^{K}$. We define C_0 as the fraction of problems where $|\mathcal{A}| > 1$, i.e., the multi-candidate rate.

2.3 Constraint Extraction

For each trace t_i, we prompt the same LLM to extract a set of FOL constraints $\mathcal{C}_i = \{c_{i,1}, \ldots, c_{i,m_i}\}$ that the answer must satisfy according to the reasoning in t_i. Constraints are expressed in a simplified FOL syntax amenable to Z3 parsing.

2.4 Deduplication and Weighting

We merge constraints across traces: $\mathcal{C} = \bigcup_i \mathcal{C}_i$, deduplicating syntactically identical constraints. Each unique constraint receives a weight proportional to the number of traces that independently produced it.

2.5 MAX-SAT Answer Selection

For each candidate answer $a \in \mathcal{A}$, we evaluate which constraints in $\mathcal{C}$ are satisfied when a is assumed correct, yielding a satisfaction score:

$$S(a) = \sum_{c \in \mathcal{C}} w_c \cdot \mathbf{1}[\text{SAT}(c \mid a)] \tag{1}$$

The selected answer is $\hat{a} = \arg\max_{a \in \mathcal{A}} S(a)$. In practice, we encode this as a weighted MAX-SAT instance and solve with Z3.

2.6 Baseline: Self-Consistency (SC)

SC selects the answer with the highest vote count: $\hat{a}_{\text{SC}} = \arg\max_{a \in \mathcal{A}} |\{i : a_i = a\}|$. SICA and SC differ only when the constraint-weighted ranking disagrees with the plurality vote.

3 Experimental Setup

Models. We evaluate five models spanning three architecture classes. **Instruction-scale**: Qwen2.5-7B-Instruct, Qwen2.5-14B-Instruct, Qwen2.5-72B-Instruct-AWQ (4-bit), LLaMA-3.1-8B-Instruct. **Full reasoning**: QwQ-32B. Default sampling: $K = 12$ traces, $T = 0.7$.

Domains. ProofWriter (deductive logic), FOLIO (first-order natural language inference), StrategyQA (commonsense reasoning, binary), ARC-Challenge (science, 4-choice), GSM8K (arithmetic, open-ended), and AIME (competition math).

Evaluation. We report exact-match accuracy for SICA and SC, the gap $\Delta = \text{acc}_{\text{SICA}} - \text{acc}_{\text{SC}}$ in percentage points, and McNemar's test p-values where sample size permits. We also report C_0 (multi-candidate rate) and mean constraint precision (fraction of extracted constraints that are discriminative).

4 Results

4.1 Main Results: SICA $\approx$ SC Across All Conditions

Table 1 summarizes accuracy for SICA and SC across all model-domain combinations. The overall pattern is clear: no condition yields a significant accuracy gain for SICA.

The largest powered test (Qwen-14B, $n = 804$ across ProofWriter, FOLIO, and StrategyQA combined) shows $\Delta = +0.3$ percentage points (76.5% vs. 76.2%), well within noise. The two results with $|\Delta| \geq 5$pp (LLaMA-8B at +5.0pp and QwQ-32B ProofWriter at −5.9pp) both have small samples ($n = 60$ and $n = 17$ respectively) with standard errors of 6–12pp, and neither approaches significance ($p = 0.29$ and $p > 0.3$).

Within the Qwen family, scaling from 7B to 72B improves the SC baseline substantially (56.7% $\rightarrow$ 83.3%) but does not change the SICA gap, which fluctuates between −1.7pp and +1.7pp. The cross-family result with LLaMA-8B ($\Delta = +5.0$pp, $p = 0.29$) is a directional noise point at $n = 60$; the analysis in the original experiment notes confirms this estimate is unreliable given the standard error.

QwQ-32B, a full reasoning model, achieves $C_0 < 5\%$ across all tested domains. With near-unanimous traces, SICA degenerates structurally to SC: there is nothing for the solver to adjudicate.

Table 1: Main results. Accuracy (%) for SICA and SC across models and domains. Δ = SICA − SC in percentage points. C_0 = multi-candidate rate (%). No result reaches $p < 0.10$. †Pilot ($n = 30$). ‡Combined across ProofWriter, FOLIO, StrategyQA.

Model	Domain	n	SICA	SC	Δ	C_0
Instruction-scale models						
Qwen2.5-14B	Combined‡	804	76.5	76.2	+0.3	46.0
Qwen2.5-14B	StrategyQA	60	56.7	53.3	+3.3	10.0
Qwen2.5-14B	ARC-Challenge	60	95.0	96.7	−1.7	5.0
Qwen2.5-14B	GSM8K†	30	86.7	86.7	0.0	13.3
Qwen2.5-7B	NLI (D012)	60	58.3	56.7	+1.7	61.7
Qwen2.5-72B	NLI (D012)	60	81.7	83.3	−1.7	38.3
LLaMA-3.1-8B	NLI (D012)	60	65.0	60.0	+5.0	60.0
Full reasoning model						
QwQ-32B	AIME	22	72.7	72.7	0.0	0.0
QwQ-32B	ProofWriter	17	58.8	64.7	−5.9	18.8

Table 2: Constraint ablation on Qwen-14B ($n = 804$). All methods perform comparably; the solver adds no information beyond constraint counting.

Method	Accuracy (%)	Δ vs. SC (pp)	Agreement w/ MAX-SAT
MAX-SAT (SICA)	76.5	+0.3	—
Count-only	76.6	+0.4	801/804
Random-weight	76.4	+0.1	799/804
SC baseline	76.2	—	—

4.2 Mechanism Analysis: MAX-SAT ≈ Counting

The constraint ablation (Table 2) provides the central mechanistic evidence. We compare three aggregation strategies applied to the same set of extracted constraints from the Qwen-14B $n = 804$ dataset:

- **MAX-SAT**: the full SICA pipeline (Z3 weighted MAX-SAT).
- **Count-only**: select the answer supported by the most constraints, ignoring satisfiability.
- **Random-weight**: assign random weights to constraints, then apply weighted MAX-SAT.

All three methods agree on 799 out of 804 problems (99.4%). MAX-SAT achieves 76.5%, count-only 76.6%, random-weight 76.4%, and the SC baseline 76.2%. No pairwise comparison is significant ($p > 0.78$ for all). On the 5 divergent problems, MAX-SAT wins 2 and loses 3 relative to count-only, showing no systematic advantage.

This result confirms that the Z3 solver's satisfiability reasoning contributes no information beyond what is already captured by constraint frequency. When constraints lack discriminative power (mean precision 0.54, barely above the chance level of 0.50), the "satisfy as many constraints as possible" objective reduces to "pick the answer with the most constraints," which in turn correlates with the plurality vote.

4.3 Temperature and Diversity Do Not Rescue SICA

One hypothesis for the null result is insufficient output diversity: perhaps higher temperature sampling would produce more candidate answers and give the solver more to work with. Table 3 tests this with Qwen-14B on NLI ($n = 60$) at $T \in \{0.3, 0.5, 0.7, 1.0\}$.

Temperature does control diversity: C_0 increases monotonically from 30% at $T = 0.3$ to 58.3% at $T = 1.0$. However, the SICA gap shows no corresponding trend. At $T = 0.3$ and $T = 0.5$, the gap is exactly 0.0pp. At $T = 0.7$, the gap is +3.3pp (the largest positive value in

Table 3: Temperature ablation, Qwen-14B, NLI ($n = 60$, $K = 12$). C_0 increases with temperature but the SICA gap remains in the noise band.

T	C_0 (%)	SICA (%)	SC (%)	Δ (pp)
0.3	30.0	75.0	75.0	0.0
0.5	40.0	73.3	73.3	0.0
0.7	53.3	73.3	70.0	+3.3
1.0	58.3	70.0	71.7	−1.7

Table 4: C_0 clusters across all model-domain conditions ($n = 12$ data points). No intermediate regime exists between the two clusters.

Cluster	C_0 **range (%)**	**Mean** C_0	**Mean** Δ **(pp)**	Δ **range (pp)**
Instruction-scale	38–62	48.2	+0.3	[−3.3, +5.0]
Reasoning / ceiling	0–5	3.1	−1.2	[−5.9, 0.0]

the study, but $n = 60$, SE $\approx$ 6pp). At $T = 1.0$, the gap reverses to −1.7pp. No temperature condition reaches $p < 0.10$.

Across the tested temperature range (0.3–1.0), increasing diversity did not translate to SICA gains, consistent with the capability trap hypothesis: diversity is a necessary but insufficient condition, and the missing ingredient (discriminative constraints) does not improve with temperature.

4.4 Operating Regime Boundaries

The C_0 diagnostic (Table 4) aggregates all model-domain conditions into a scatter of C_0 vs. Δ. Two clusters emerge with no intermediate points:

- **Instruction-scale cluster** ($C_0 = 38$–62%, $n = 8$ conditions): Mean gap +0.3pp, range $[-3.3, +5.0]$pp. High diversity, but constraints are non-discriminative. SICA $\approx$ SC.
- **Reasoning/ceiling cluster** ($C_0 < 5\%$, $n = 4$ conditions): Mean gap −1.2pp. Models are too consistent for symbolic aggregation to add anything. SICA $\equiv$ SC structurally.

The Pearson correlation between C_0 and Δ is $r = 0.08$ ($p = 0.81$) across the 12 data points. C_0 does not predict the direction or magnitude of the gap. Its value is purely as a binary gate: $C_0 < 5\%$ reliably indicates that SICA will degenerate to SC (all such conditions show $\Delta \leq 0$), allowing practitioners to skip the overhead without risk.

4.5 Comparison with Alternative Aggregation Methods

Weighted SC. Table 5 compares four trace-weighting schemes applied to the Qwen-14B $n = 804$ dataset. Inverse-length weighting achieves the highest accuracy (76.7%), outperforming SC by 0.5pp ($p = 0.45$), but no method differs significantly from SC or from SICA. The result confirms that more sophisticated aggregation, whether symbolic or heuristic, provides no reliable gain over uniform voting.

Logic-LM $\times$ K. We also compare against a Logic-LM style baseline on FOLIO ($n = 204$), where each trace undergoes independent FOL verification before majority voting over verified traces. Logic-LM $\times$ K achieves 78.9%, compared to SC at 79.4% and SICA at 79.4%. FOL verification rejects 28% of traces, but 31% of rejected traces are actually correct, introducing verification noise that offsets any filtering benefit. Symbolic methods face limitations at both the aggregation level (SICA) and the per-trace verification level (Logic-LM).

Table 5: Weighted SC variants on Qwen-14B ($n = 804$). No weighting method significantly outperforms uniform SC.

Method	Accuracy (%)	Δ vs. SC (pp)	p-value
SC (uniform)	76.2	—	—
Inverse-length	76.7	+0.5	0.45
Length-weighted	75.3	−0.9	0.15
Constraint-count	75.3	−0.9	0.20
Sica (MAX-SAT)	76.5	+0.3	—

Table 6: Tiebreak analysis on the $n = 804$ Qwen-14B dataset. Sica is an effective tiebreaker, but ties are too rare (5.3%) to impact overall accuracy.

Tiebreak method	Accuracy on ties (%)	Overall accuracy (%)
Sica (MAX-SAT)	78.1	76.5
Random	50.0	75.6
Alphabetical	59.4	75.9
First-seen	43.8	75.5

4.6 Tiebreak Quality: Locally Effective, Globally Negligible

Sica is a high-quality tiebreaker on the rare occasions when SC produces a tie (Table 6). Among the 32 tied problems in the $n = 804$ dataset (5.3% of all problems), Sica selects the correct answer 78.1% of the time, compared to 50.0% for random tiebreaking ($p = 0.031$). However, because ties affect only 5.3% of problems, the overall accuracy impact is less than 1 percentage point (76.5% vs. 75.6% for SC with random tiebreaking). The K ablation experiment (Appendix A) further shows that at odd K (e.g., $K = 11$), where ties are structurally rare, the Sica gap vanishes entirely ($\Delta = 0.0$pp), confirming that tiebreaking accounts for the method's entire (small) contribution.

4.7 Computational Overhead

Table 7 breaks down wall-clock time per problem. Sica adds approximately 48% overhead relative to SC, with constraint extraction (an additional LLM call per trace) accounting for nearly all of the added cost. The MAX-SAT solver itself is negligible ($< 0.01\%$ of total time). For a method that yields $\Delta \approx 0$pp, this overhead is difficult to justify.

5 Discussion

5.1 The Capability Trap

Our results point to a consistent pattern across all tested conditions: symbolic consensus methods face an empirically pervasive limitation. The two requirements for symbolic aggregation to outperform voting (output diversity and discriminative constraints) are not simultaneously met in any configuration we tested.

Instruction-scale models (7B–72B) produce diverse outputs ($C_0 = 38$–62%), giving the solver multiple candidates to adjudicate. But the constraints extracted from their reasoning traces have precision near chance (0.54 across 307 multi-candidate problems, $\sigma = 0.21$). When binned by vote entropy, aggregate precision shows a weak directional trend (0.51, 0.54, 0.57 from low to high entropy) but with overlapping 95% confidence intervals. Per-problem noise is substantial ($R^2 = 0.008$ for the raw precision-entropy scatter), though the aggregate direction is consistent across all three domains tested (ProofWriter: 0.54, FOLIO: 0.52, StrategyQA: 0.55).

Table 7: Per-problem wall-clock breakdown ($n = 804$). Constraint extraction nearly doubles the cost relative to SC, while the solver is instantaneous.

Stage	Mean time (s)	% of total
Trace generation	32.0	51.6
Constraint extraction	29.0	46.8
Deduplication	0.9	1.5
MAX-SAT solving	0.004	< 0.01
Total (SICA)	62.0	100.0
Total (SC only)	32.0	—
SICA overhead vs. SC	+30.0	+48.4%

Full reasoning models (QwQ-32B) take a different path to the same outcome. Their chain-of-thought reasoning is sufficiently deterministic that even at $T = 0.7$–1.0, 12 sampled traces converge to the same answer on $> 95\%$ of problems. With no diversity, there is nothing for the solver to resolve.

Qualitative analysis of 8 representative problems confirms the mechanism. Of the constraints extracted, 85% are syntactically valid and 62% are semantically relevant to the question, but only 18% are discriminative (i.e., satisfied by the correct answer but not by an incorrect one). When constraints cannot distinguish answers, MAX-SAT's optimization reduces to counting, as confirmed by the ablation in Section 4.2.

5.2 C_0 as a Binary Early-Stopping Gate

Rather than a fine-grained diagnostic predictor, C_0 is useful as a binary decision rule. Our data support a simple recommendation: if $C_0 < 5\%$ after sampling K traces, skip symbolic aggregation entirely and use SC. All conditions with $C_0 < 5\%$ show $\Delta \leq 0$pp, meaning SICA never helps and occasionally hurts (e.g., QwQ-32B on ProofWriter, $\Delta = -5.9$pp at $n = 17$).

For conditions with $C_0 > 30\%$, the picture is less clear. The gap ranges from -3.3pp to $+5.0$pp with no predictable direction ($r = 0.08$, $p = 0.81$). In this regime, SICA is not reliably better or worse than SC, but it incurs 48% computational overhead. The practical recommendation is the same: use SC.

5.3 Implications and Limitations

Our findings do not imply that symbolic reasoning is fundamentally incompatible with LLM outputs. They show that the specific pipeline of constraint extraction followed by MAX-SAT aggregation does not improve over voting with current models and extraction methods. Future work on improved constraint extraction (e.g., via specialized verifiers or interactive refinement) may change this calculus.

Several limitations apply. First, all instruction-scale results use $n \leq 804$ per condition; larger samples might reveal small but real effects below our detection threshold. Second, our constraint extraction uses the same LLM that generates the traces, which may limit constraint quality. Third, we test a single symbolic aggregation framework (SICA); alternative formalisms (e.g., probabilistic logic, constraint relaxation) may behave differently.

6 Related Work

Self-Consistency and voting methods. Self-Consistency [Wang et al., 2023] established majority voting over sampled reasoning paths as a strong baseline for LLM reasoning. Universal Self-Consistency [Chen et al., 2023] extends this to free-form answers via LLM-based answer

clustering. Our work takes SC as the baseline and asks whether symbolic methods can improve upon it.

Neuro-symbolic reasoning. Logic-LM [Pan et al., 2023] translates NL problems into symbolic representations and uses logic solvers for inference. LINC [Olausson et al., 2023] generates FOL translations for natural language inference. These methods apply symbolic reasoning to individual problems; our work tests whether symbolic methods help when aggregating across multiple sampled traces.

Verification and reranking. Outcome reward models (ORM) and process reward models (PRM) [Cobbe et al., 2021, Lightman et al., 2023] train separate verifiers to score and rerank reasoning traces. These approaches require training data and a separate model. SICA attempts to achieve similar selection quality using symbolic constraints extracted from the traces themselves, without additional training.

Constraint satisfaction in NLP. MAX-SAT and related constraint satisfaction methods have been applied to various NLP tasks including semantic parsing, information extraction, and structured prediction. Our work applies weighted MAX-SAT specifically to the answer selection problem in multi-trace LLM reasoning.

7 Conclusion

We presented a systematic empirical study of symbolic consensus for LLM reasoning, evaluating SICA (MAX-SAT over extracted FOL constraints) against majority voting across 5 models, 6 domains, and over 1,300 problems. The method provides no significant accuracy gain in any tested condition. The constraint ablation reveals why: extracted constraints lack discriminative power, causing MAX-SAT to degenerate to constraint counting. We characterize this as a capability trap that is empirically pervasive across the tested conditions and model scales. For practitioners considering symbolic aggregation, we recommend $C_0 < 5\%$ as a zero-cost early-stopping gate, and SC as the default aggregation strategy until constraint extraction methods improve substantially.

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A K Ablation

Table 8 shows the effect of ensemble size K on SICA vs. SC for Qwen-14B on NLI ($n = 60$). The gap is insensitive to K: no value from $K = 6$ to $K = 20$ achieves significance. At $K = 11$ (odd), the gap is exactly 0.0pp, confirming that SICA's only mechanism is tiebreaking in even-K scenarios. At large K (≥ 16), SICA is slightly worse than SC ($\Delta = -1.7$pp), consistent with noise injection from additional low-quality constraints.

Table 8: K ablation, Qwen-14B, NLI ($n = 60$). SICA is insensitive to K; the $K = 11$ (odd) condition confirms the tiebreak-only mechanism.

K	SICA (%)	SC (%)	Δ (pp)
6	73.3	73.3	0.0
8	71.7	73.3	−1.7
11	73.3	73.3	0.0
12	73.3	70.0	+3.3
16	71.7	73.3	−1.7
20	70.0	71.7	−1.7

B Constraint Quality by Domain

Table 9 shows per-domain constraint precision across the 307 multi-candidate problems in the Qwen-14B dataset. All domains have mean precision near 0.53–0.55, with standard deviation ≈ 0.21. The consistency of this near-chance precision across structurally different domains (formal logic, commonsense, natural language inference) suggests a general limitation of LLM-based constraint extraction rather than a domain-specific artifact.

Table 9: Constraint precision by domain (multi-candidate problems only, Qwen-14B).

Domain	Mean precision	n (problems)
ProofWriter	0.54	—
FOLIO	0.52	—
StrategyQA	0.55	—
Overall	0.538 ($\sigma = 0.21$)	307

C Overhead by Domain

Extraction overhead varies modestly by domain: ProofWriter incurs 51% overhead relative to SC, while FOLIO incurs 39%. The difference reflects trace length variation (ProofWriter traces are typically longer and require more extraction effort). In all cases, the overhead is substantial relative to the null accuracy gain.

D Within-Family Scaling: Qwen 7B/14B/72B

Table 10 shows the Qwen scaling trajectory. SC baseline improves dramatically (56.7% $\rightarrow$ 83.3%) as model size increases. C_0 decreases (61.7% $\rightarrow$ 38.3%) as stronger models become more consistent. The SICA gap remains in the noise band at all scales, ranging from -1.7 to $+1.7$pp. Constraint precision is flat (0.51–0.54), confirming that scaling alone does not improve constraint discriminability.

Table 10: Qwen family scaling on NLI ($n = 60$, $K = 12$, $T = 0.7$). Stronger models improve SC but not the SICA gap.

Model	**SC (%)**	**SICA (%)**	**Δ (pp)**	C_0 **(%)**	**Precision**
Qwen2.5-7B	56.7	58.3	+1.7	61.7	0.51
Qwen2.5-14B	70.0	73.3	+3.3	53.3	0.53
Qwen2.5-72B	83.3	81.7	-1.7	38.3	0.54

E Logic-LM Baseline Details

On FOLIO ($n = 204$), the Logic-LM $\times$ K baseline applies per-trace FOL verification followed by majority voting over verified traces. Of 28% rejected traces, 31% were actually correct, yielding a net accuracy of 78.9% compared to SC at 79.4%. This confirms that symbolic verification noise is not limited to the aggregation stage: per-trace FOL verification is also unreliable with current LLM-generated formalizations.

F GSM8K Pilot

A pilot evaluation on GSM8K ($n = 30$, Qwen-14B) shows $C_0 = 13.3\%$ and $\Delta = 0.0$pp. The model's high baseline accuracy (86.7%) leaves few multi-candidate problems. Mathematical constraints have slightly higher precision (0.58) than NLI domains, but the sample is too small for reliable conclusions. This pilot extends domain coverage but should not be interpreted as evidence for or against SICA in the mathematics domain.